



Calculus Honors

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · GRADES 9–12 · FULL YEAR · HONORS

~4¼ hrs per week · 50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in lesson order, activities, and examples from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Calculus Honors develops a scholar's understanding of change, rate, and accumulation — the fundamental tools for studying motion and transformation. The year builds rigorously: limits and continuity first, then derivatives applied to slopes, motion, related rates, and optimization. Careful study of how limits and derivatives reveal function behavior makes graphing and analysis genuinely deep. The year closes with integrals — areas, volumes, average values — and differential equations. VR and AI enhancements support visualizing rates of change and accumulation. Throughout, scholars develop precision in symbolic reasoning, rigor in handling infinite processes, and care in connecting abstract mathematics to the phenomena it models. Prerequisite: Precalculus or Probability and Statistics.

THE YEAR AT A GLANCE

1 Limits and Continuity (foundation)

The rigorous idea underneath all of calculus: what it means to approach, and when functions behave.

3 Analyzing Functions (major unit)

Using limits and derivatives to reveal function behavior — graphing and analysis with depth.

5 Differential Equations (major unit)

Equations whose unknowns are functions — solved with the year's full toolkit.

2 Derivatives (major unit)

Finding and applying derivatives: slopes, motion, related rates, and optimization.

4 Integrals (major unit)

Accumulation made precise: evaluating integrals and applying them to areas, volumes, and average values.

The Honors difference. Rigor in handling infinite processes and precision in symbolic work throughout — genuine preparation for college mathematics and the sciences.

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- Visualization work with VR and AI tools makes rates of change and accumulation visible; quizzes and tests close each unit.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A graphing calculator — limits, derivatives, and integrals lean on it.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.